

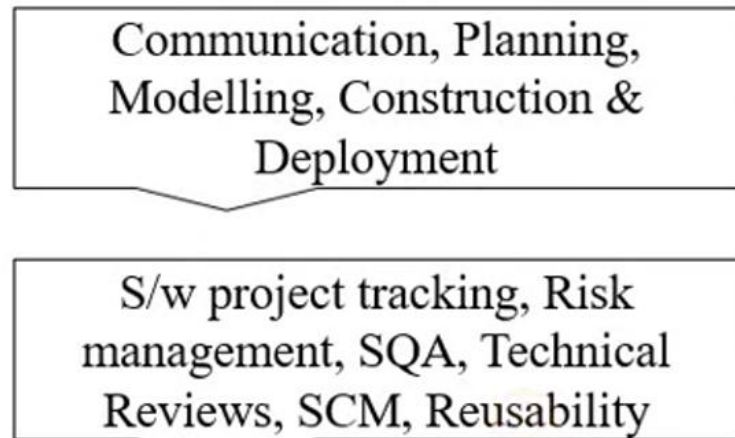
Unit 1 Process Model



Software Process

- A software process is the set of activities and associated outcome that produce a software product.
 - Software engineers mostly carry out these activities.
1. **Software Specifications:** The functionality of the software and constraints on its operation must be defined.
 2. **Software Development:** The software to meet the requirement must be produced.
 3. **Software Validation:** The software must be validated to ensure that it does what the customer wants.
 4. **Software Evolution:** The software must evolve to meet changing client needs.

Generic Process Model



- **Software process** means to develop quality products by using technical & management rules.
- **The generic process model** is an description of the software development process.
- It is used in most software models since it provides a base for them.

Software process

Process framework

Umbrella activities

framework activity # 1

software engineering action #1.1

Task sets

⋮

software engineering action #1.k

Task sets

work tasks
work products
quality assurance points
project milestones

work tasks
work products
quality assurance points
project milestones

⋮

framework activity # n

software engineering action #n.1

Task sets

⋮

software engineering action #n.m

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work tasks
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project milestones

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Process Framework Activities

1. Communication:

- The software development starts with the communication between customer and developer.
- Developer gather requirements of the project with the users.
- Communication with consumers and stakeholders to determine the system's objectives, software's requirements & features.

2. Planning:

- It consists of complete estimation, scheduling for project development and tracking.
- Discuss work plan, technical risk, list of resource requirements, work produced & work schedule.

3. Modelling:

- Develop a practical model to get a better understanding of the project.
- It consists of complete requirement analysis and design of the project like algorithm, flowchart.
- The algorithm is the step-by-step solution of the problem.
- The flow chart shows a complete flow diagram of a program.
- If changes are required, we implement them in this step.

4. Construction:

- Construction consists of code generation and the testing part.
- Coding part implements the design details using an appropriate programming language.
- Testing is to check whether the flow of coding is correct or not.
- Testing also check that the program provides desired output.

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Umbrella Activities



➤ Activities that occur throughout a software process for better management and tracking of the project.

1. Software Project Tracking and Control:

- Compare the progress of the project with the plan and take steps to maintain a planned schedule.



2. Risk Management:

- Analyze any potential risks that may have an impact on the software product's quality and outcome.

3. Software Quality Assurance (SQA):

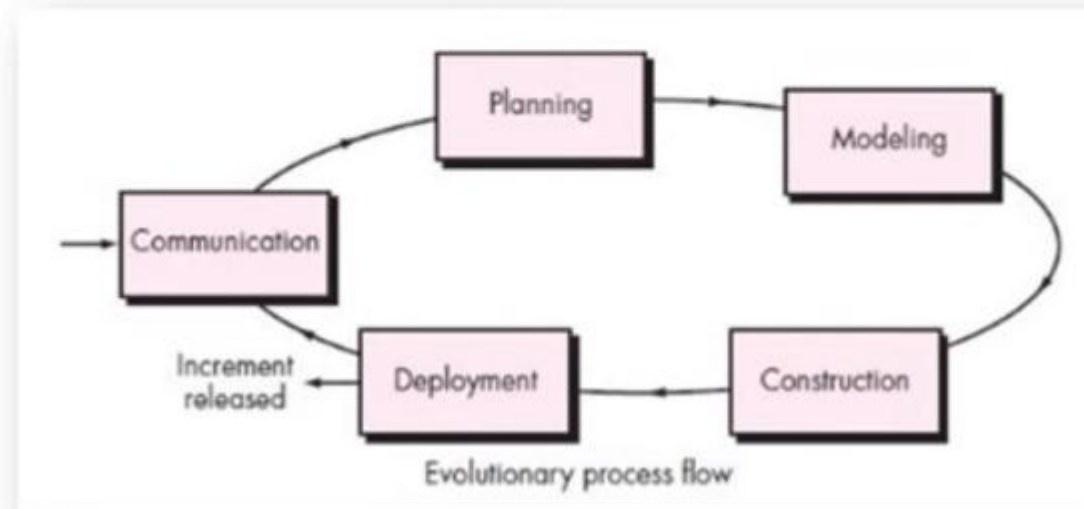
- These are activities required to maintain software quality.
- Perform actions to ensure the product's quality.



Process Framework Activities

5. Deployment:

- Deployment step consists of delivering the product to the customer and take feedback from them.
- If the customer wants some corrections or demands for the additional capabilities, then the change is required for improvement in the quality of the software.



Umbrella Activities

4. Technical Reviews :

- Assessment of errors and correction done at each stage of activity.

5. Software Configuration Management (SCM):

- Managing of configuration process when any change in the software occurs.

6. Reusability Management:

- Reusable work items should be backed up, and reusable software components should be achieved.

7. Work Product Preparation and Production:

- The activities to create models, documents, logs, forms and lists are carried out.

Framework Activities

➤ Consider Communication Activity

for small project:

- Making a phone call with stakeholder.
- Discuss requirements and note them down.
- Organize requirements.
- Mail stakeholder for review and approval.

➤ Consider Communication Activity

for large project:

- Arrange live meeting
- Complete feasibility study
- Requirement analysis
- Specification documents

Software process

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Scope of Project

- **Project Scope** defines the **boundaries** of a project — what will be done, what will not be done, who is responsible, the time frame, budget, and deliverables. It helps keep the project focused and prevents "scope creep" (uncontrolled changes or expansion).
- **Project Title:**
Website Development for **SSN College**
- **Goal:**
Design and develop a modern, responsive, and user-friendly website for SSN College to improve communication with students, faculty, and prospective applicants.

✓ In Scope

- Design and development of main website
- Pages: Home, About Us, Courses, Admissions, Events, Faculty Directory, Contact
- Online admission form (PDF upload)
- Responsive layout for desktop and mobile
- Content Management System (WordPress or similar)
- SEO (Search Engine Optimization) setup for Google indexing

✗ Out of Scope

- Development of a student login portal
- Integration with fee payment systems
- Mobile app creation

📄 Assumptions

- SSN College will provide all written content and images
- Hosting and domain are already managed by SSN IT department

⚠ Constraints

- Must be completed within 8 weeks, before the admission cycle starts
- Budget capped at ₹6,50,000

5 W's for SSN College Website Project

W	Explanation	SSN College Example
Who	Stakeholders involved	Web development agency, SSN IT team, admin staff, faculty, students, prospective applicants
What	Project deliverables	A fully functional college website with academic and admission information
When	Timeline and deadlines	Start on August 1 , finish by September 25 (8 weeks)
Where	Delivery and accessibility	Hosted on www.ssn.edu.in , accessible globally on desktop and mobile
Why	Business or institutional need	To modernize digital presence, support online admissions, and improve stakeholder communication

Process Assessment and Improvement

- Software Process (SP) cannot guarantee that software will be delivered on time, meet the needs, or has the desired technical characteristics.
- **CMM levels are stages of process maturity:**

Initial: Ad hoc and chaotic processes. (lack formal structure or consistency)

Managed: Project management and documentation.

Defined: Standardized and documented processes.

Quantitatively Managed: Performance measured and controlled.

Optimizing: Continuous process improvement and innovation.

- **Standard CMMI Assessment Method for Process Improvement (SCAMPI)**
— provides a five step process assessment model that incorporates five phases: initiating, diagnosing, establishing, acting and learning.
- **CMM-Based Appraisal for Internal Process Improvement (CBA IPI)**—provides a diagnostic technique for assessing the relative maturity of a software organization; uses the SEI CMM as the basis for the assessment.
- **SPICE—The SPICE** (ISO/IEC15504) standard defines a set of requirements for software process assessment. The intent of the standard is to assist organizations in developing an objective evaluation of the efficacy of any defined software process. [ISO08]. **Software Process Improvement and Capability dEtermination,**
- **ISO 9001:2000 for Software—a generic standard that applies to any organization**

that wants to improve the overall quality of the products, systems, or services that it provides. Therefore, the standard is directly applicable to software organizations and companies



Software Engineering Paradigm

- Software paradigm or process model is an abstract representation of a process. The process model is chosen based on nature of software project and application and then to obtain deliverable product method and tools are applied

integration which delivers the results

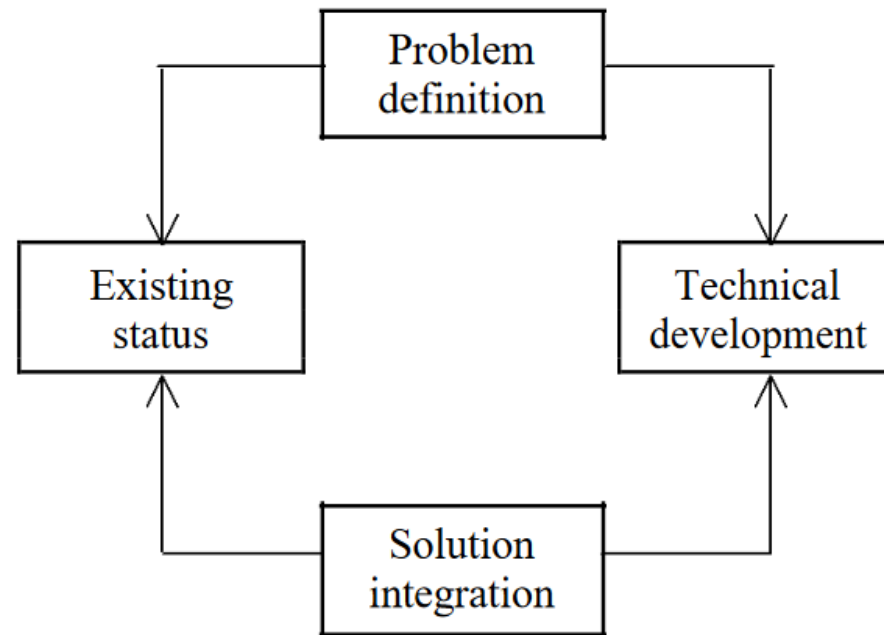


Fig. 1.5 The Phase of a Problem Solving Loop

Need for Life Cycle Model

- Structured Approach:** It provides a systematic framework for organizing and managing project phases from conception through to retirement.
- Risk Management:** By breaking down the project into manageable stages, potential risks can be identified and mitigated early.
- Resource Allocation:** Helps in planning and allocating resources effectively throughout the project.
- Quality Control:** Establishes clear milestones and deliverables, improving the ability to monitor and ensure quality.
- Communication:** Facilitates clear communication among stakeholders by defining phases, roles, and responsibilities.
- Consistency:** Promotes consistency and standardization in processes, leading to more predictable outcomes and easier project management.

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- i) Waterfall Model
 - (ii) Prototyping Model
 - (iii) Incremental Model
 - (iv) Spiral Model
 - (v) WIN WIN Spiral Model
 - (vi) Evolutionary Model
 - (vii) Object Oriented Model

Waterfall Model Design

Key Characteristics:

- **Linear and Sequential:** Each phase flows into the next without overlap.
- **Document-Driven:** Extensive documentation is produced at each stage.
- **Rigid:** Changes are difficult to implement once a phase is completed.
- **Predictable:** Clear milestones and deliverables at each phase.

Suitable For:

- Projects with well-understood and unchanging requirements.
- Smaller projects with limited complexity.
- Projects where a structured approach is required, such as those in regulatory environments.

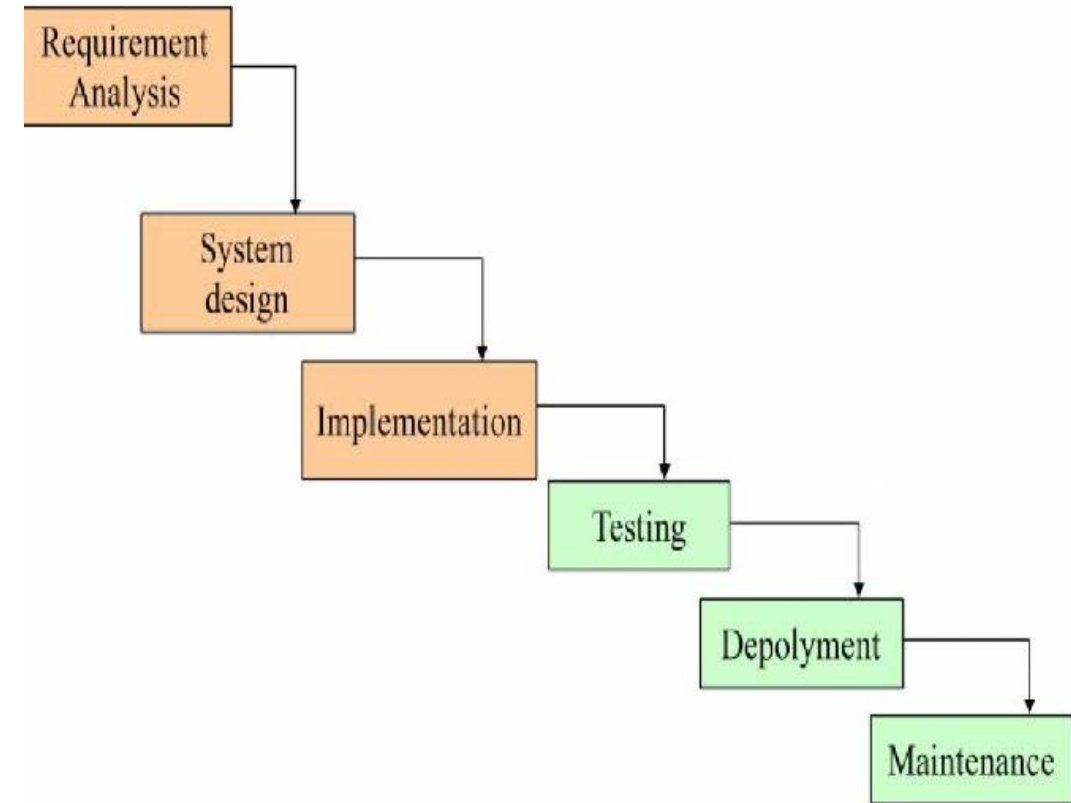


Fig. 1.6 Different Phases in Waterfall Model

Pros	Cons
• Simple and easy to understand and use • Easy to manage due to the rigidity of the model . each phase has specific deliverables and a review process. • Phases are processed and completed one at a time. • Works well for smaller projects where requirements are very well understood. • Clearly defined stages. • Well understood milestones. • Easy to arrange tasks. • Process and results are well documented.	• No working software is produced until late during the life cycle. • High amounts of risk and uncertainty. • Not a good model for complex and object-oriented projects. • Poor model for long and ongoing projects. • Not suitable for the projects where requirements are at a moderate to high risk of changing. So risk and uncertainty is high with this process model. • It is difficult to measure progress within stages. • Cannot accommodate changing requirements. • No working software is produced until late in the life cycle. • Adjusting scope during the life cycle can end a project. • Integration is done as a “big-bang. at the very end, which doesn’t allow identifying any technological or business bottleneck

V-Model

- The V - model is SDLC model where execution of processes happens in a sequential manner in V-shape. It is also known as Verification and Validation model. extension of the waterfall model

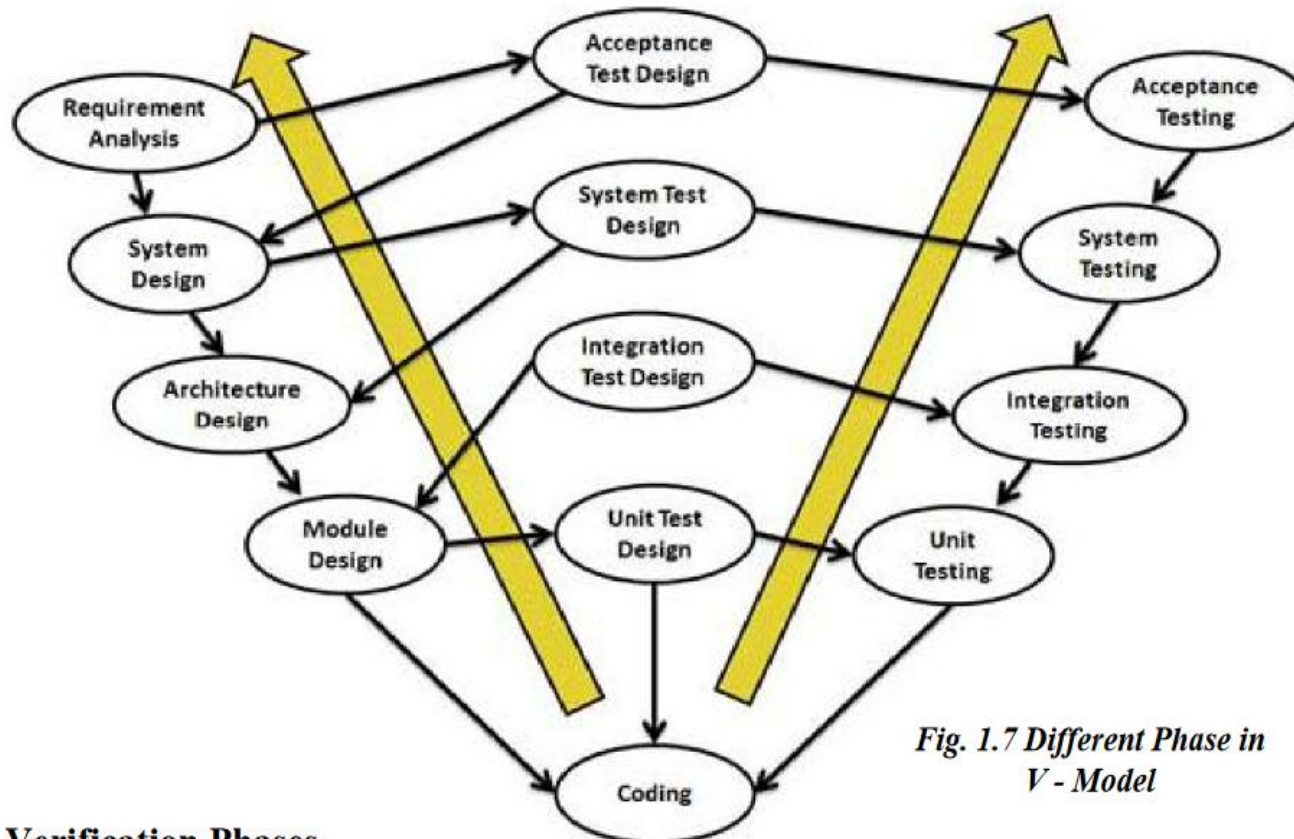


Fig. 1.7 Different Phase in V - Model

- **Development Phases:** On the left side of the "V," these phases progress sequentially, similar to the Waterfall Model.
 - **Requirements Analysis:** Understanding what the system should do.
 - **System Design:** Creating an overall system architecture.
 - **High-Level Design:** Detailing subsystems and interfaces.
 - **Low-Level Design:** Designing individual components.
 - **Implementation:** Writing the actual code.
- **Testing Phases:** On the right side of the "V," each testing phase corresponds to a development phase.
 - **Unit Testing:** Tests individual components against the low-level design.
 - **Integration Testing:** Ensures that subsystems work together, based on the high-level design.
 - **System Testing:** Tests the complete system against the system design.



Key Characteristics:

- **Parallel Development and Testing:** Emphasizes that each development stage has a corresponding testing stage.
- **Early Detection of Defects:** Issues can be identified and addressed early due to the parallel nature of development and testing.
- **Verification and Validation:** Ensures both that the system is built correctly (verification) and that the right system is built (validation).

Suitable For:

- Projects where requirements are well-understood and unlikely to change.
- Environments requiring a high level of documentation and traceability, such as aerospace, defense, and medical industries.



Pros	Cons
• This is a highly disciplined model and Phases are completed one at a time. • Works well for smaller projects where requirements are very well understood. • Simple and easy to understand and use. • Easy to manage due to the rigidity of the model each phase has specific deliverables and a review process.	• High risk and uncertainty. • Not a good model for complex and object-oriented projects. • Poor model for long and ongoing projects. • Not suitable for the projects where requirements are at a moderate to high risk of changing. • Once an application is in the testing stage, it is difficult to go back and change a functionality • No working software is produced until late during the life cycle.